Ex No: 10

Date: 10.9.26

Interfacing with Routers in Cisco Packet Simulator.

a) Simple Internetwork using a Router

Aim:

To Design and configure a simple internetwork using a router in Cisco Packet Tracer and observe the routing of packets between networks.

Procedure:

1) Open Cisco Packet Tracer.
2) Place on Cisco 2911 Router, two switches and PCs for each network.
3) Connect the PCs the switches using copper straight cables.
4) Connect the switches to the router using Ethernet cables.
5) Assign static IP addresses to all PCs and router interface.
6) Configure the router interface using assigned IP address.
7) Enable the router interface using no shutdown command.
8) Verify the interface status using the IP interface brief.
9) Using simple PDU to send packet from one network to other.
10) Select Simulation mode and observe the packet delivery
11) Observe that the router forwards the packet through the appropriate interface according to destination network.

Result:

The simple internetwork successfully design and configured using a router in CPT. Packet transmission and routing between different networks was successfully.

5) Wireless Router with DHCP server and Internet cloud.

Aim:

1) Open Cisco Packet Tracer.

2) Place one cloud-PT, one WRT 300N wireless Router, one PC and two laptop.

3) Connect the cloud-PT to the WAN port of WRT 300N using copper straight through cable.

4) Connect the PC to a LAN port of the WRT 300N using a copper straight LAN cable.

5) Install a wireless such wireless in the laptops if required.

6) Configure the WRT 300 N with LAN IP address 192.169.01 and Subnet Mask 255.255.255.0.

7) Enable the DHCP server in wireless router.

8) Connect the laptops to the received IP address automatically.

9) Verify that the devices have received IP address successfully

10) Use the ping command to test connect between the devices.

11) Use simulation mode to observe the packet transmission through the wireless router.

Result:

The internetwork using a wireless router, DHCP server and Internet cloud was successfully designed and configured. The devices obtained IP config automatically through DHCP and communication b/w the devices was successfully verified.

6) Interneteworks using Switch, Router and Ethernet Cloud.

Aim:

To Design and configure an interface using Switch router and cables in Cisco Packet Tracer.

Procedure:

1) Open Cisco Packet Tracer.
2) Place one Cisco 2911 Router one Switch and 3PCs.
3) Connect the router to the switch using a copper straight through cables.
4) Connect the three PCs to different Ports of the Switch using straight through cables.
5) Assign an IP address to the router interface.
6) Assign an IP address to all PCs.
7) Configure the default gateway of each PC using Router.
8) Enable the router using the no shutdown IP address
9) Check the router interface status using Show command.
10) Test Connectivity between PCs using Ping Commands
11) Using Simulate mode and observe the packet travelling through Switch and router.

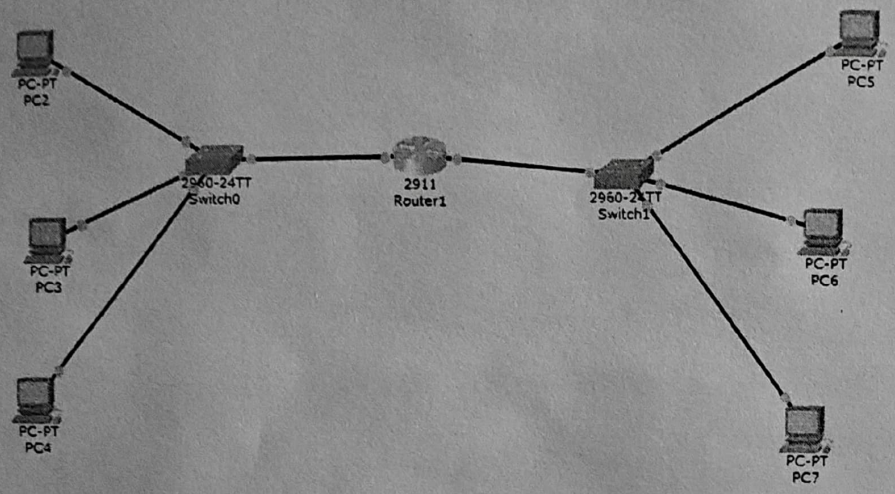
Result:

The Internetwork using a router, Switch and Ethernet Cables was succesfully designed and config. & DT communicated between the connect device was Succesfully verified using the Ping Command.

```
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

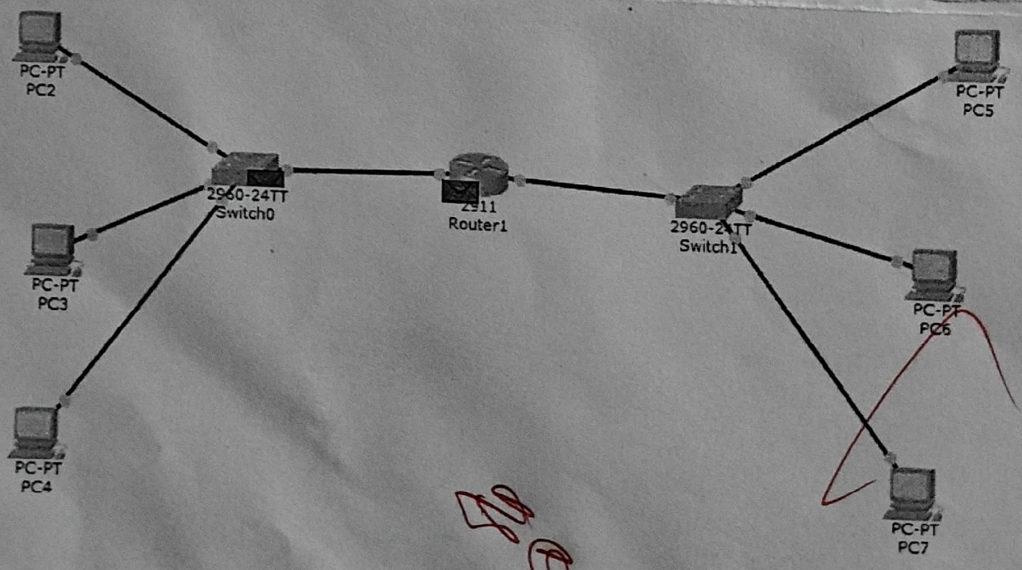
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

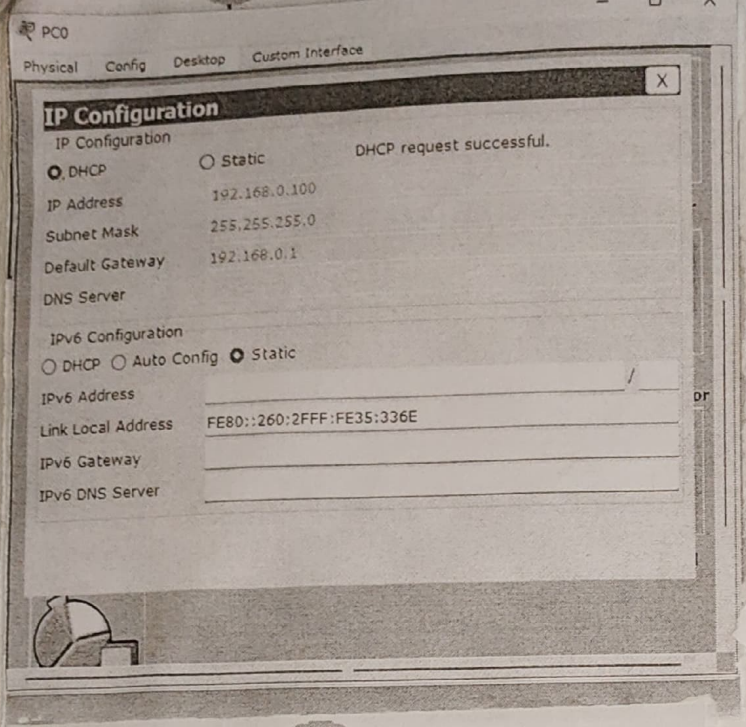
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```

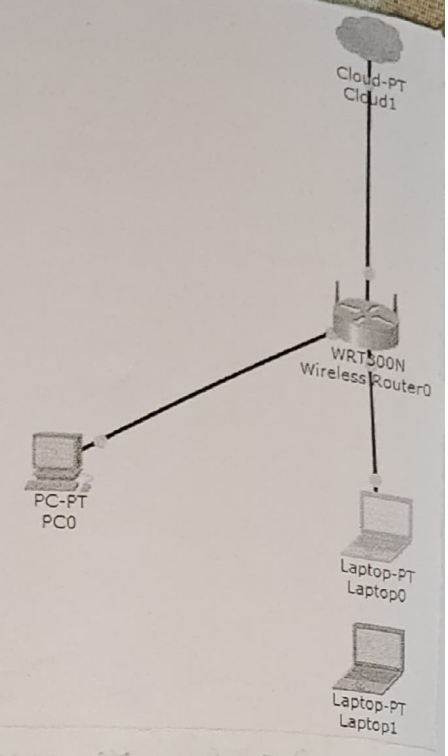


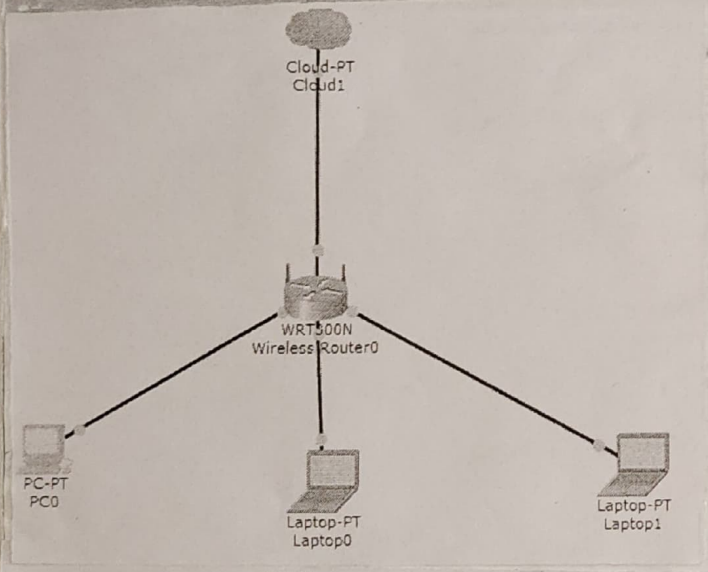

```
Router#enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

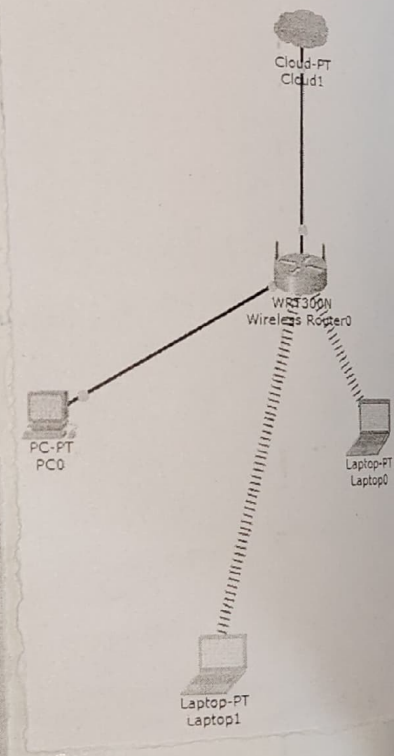
Copy Paste

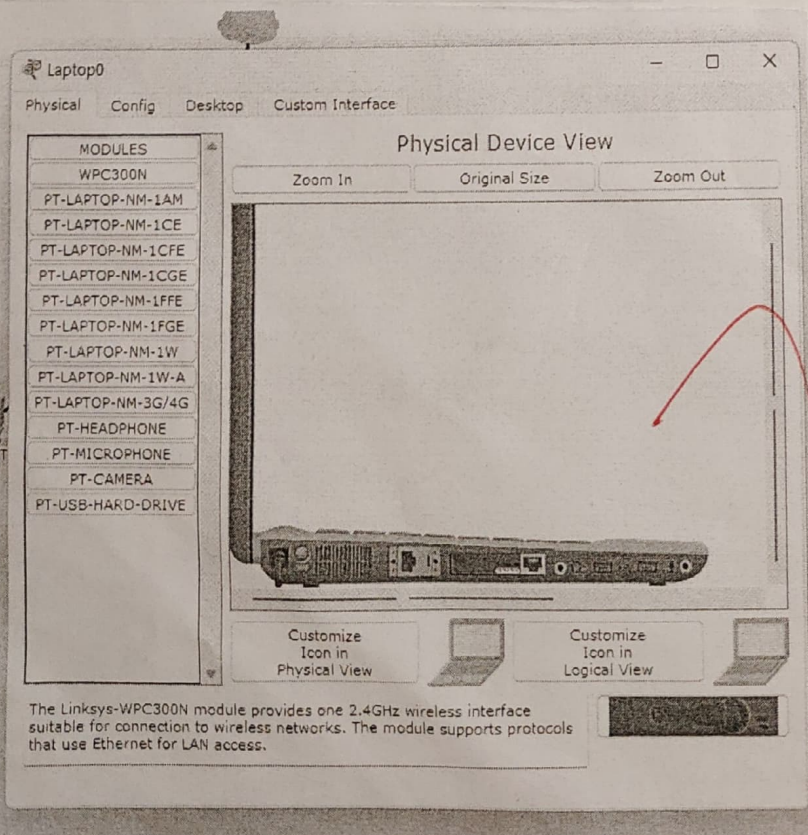



PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.0.100
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FE35:336E
IPv6 Gateway
IPv6 DNS Server


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Laptop0
Physical
Config
Desktop
Custom Interface
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
PT-CAMERA
PT-USB-HARD-DRIVE
Physical Device View
Zoom In
Original Size
Zoom Out
Customize
Icon in
Physical View
Customize
Icon in
Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.

PC0

Physical Config Desktop Custom Interface

Command Prompt

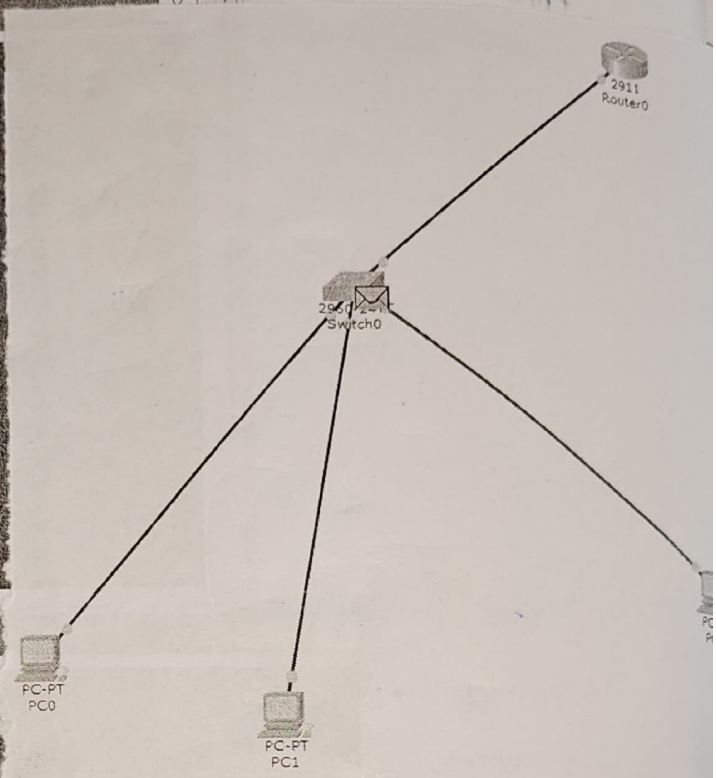
```
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>
```



Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```